# SortBuddy scan station: how to use this template



## Print

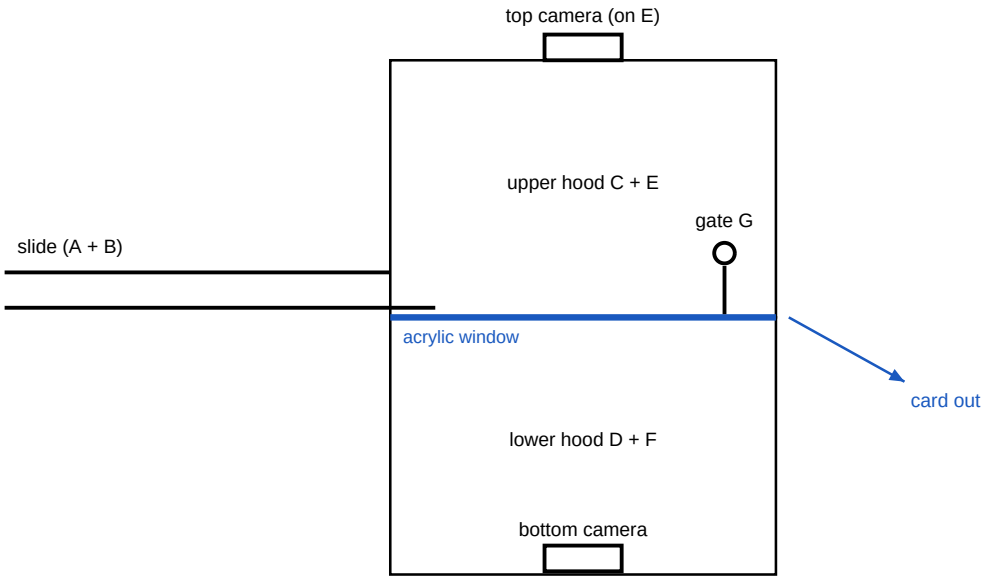
Print at 100% / 'Actual size' (NOT 'Fit to page'). Measure the 50 mm square: if it isn't 50 mm, fix the print setting.

## Cut

Tape each page lightly onto the foam board, cut through paper and foam with a FRESH blade, 2-3 light passes.
Black solid lines = cut. Crossed boxes = cut out (holes). Dashed = guides, don't cut.
Notches are exactly 5 mm (the board thickness): cut on the inside of the line so the tabs fit snug.

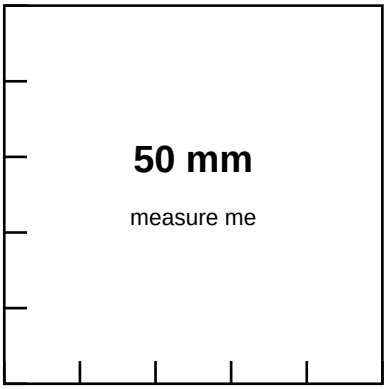
## Assemble (glue only on the OUTSIDE; nothing sticky where cards slide)

1. Lower hood: panels D1 + D2 stand upright, bottom plate F's tabs in their notches.
2. Window: acrylic 80 x 130 across the top edges of D (thin double-sided tape on those edges).
3. Upper hood: panels C1 + C2 on the acrylic, right above D; top plate E's tabs in their notches.
   Splices H (x4) on the outside across the C/D seam hold the hoods and the acrylic together.
4. Slide: walls B onto floor A (tabs in notches). Tape or plastic sheet on the floor's top.
   Its end slides between C1 and C2 onto the acrylic (15 mm). Joiners J outside, wall to panel.
5. Gate: servo through C1's cutout from outside, horn inside. Flap G glued to the horn.
   Set 'closed' in the control panel so the flap's edge just touches the window.
6. Cameras: top board on E, lens down through the hole; bottom board on F, lens up.
   Long side of the picture along the channel. Hot glue the board corners only, never the lens.
7. Stand: wedges K on base L, lower hood on the wedges, tall end under the slide.
8. Light: LED strip inside the upper hood around the camera, covered with tissue or paper.
   In the lower hood put it along the side walls, not next to the camera (glare off the acrylic).
9. Hopper (pages 9-10): walls P2-P5 around floor P1. Motor on bracket Q under the floor, wheel W (rubber bands round its rim) poking ~1 mm up through the slot. Strip T on the front wall sets the gap to ~1.5 cards. The floor's lip rests 20 mm onto the top of the slide.




Side view, not to scale. The whole thing stands tilted 30° on wedges K.

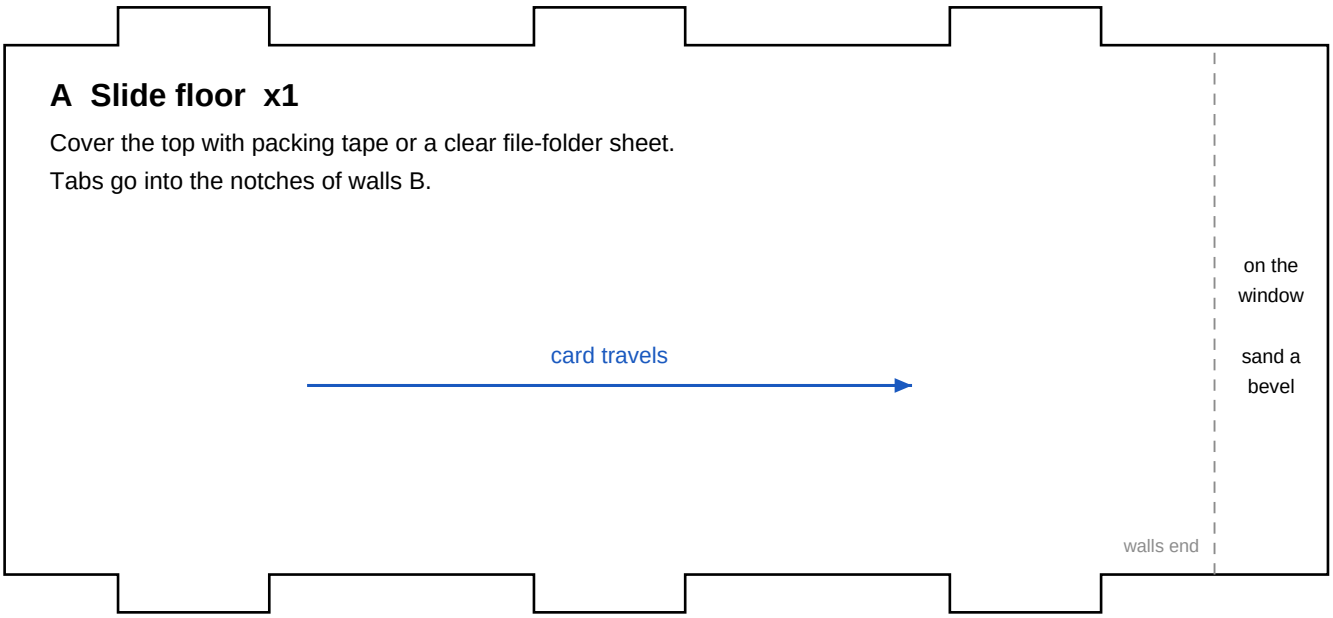
**Parts (5 mm foam):**
A slide floor x1
B slide walls x2
C1, C2 upper hood
D1, D2 lower hood
E top plate
F bottom plate
G gate flap
H splices x4
J joiners x2
K wedges x2
L base
M, N sorter (later)
P1-P5 hopper
Q motor bracket
R braces x2
S weight, W wheel x2
+ acrylic 80 x 130
+ plastic strip T

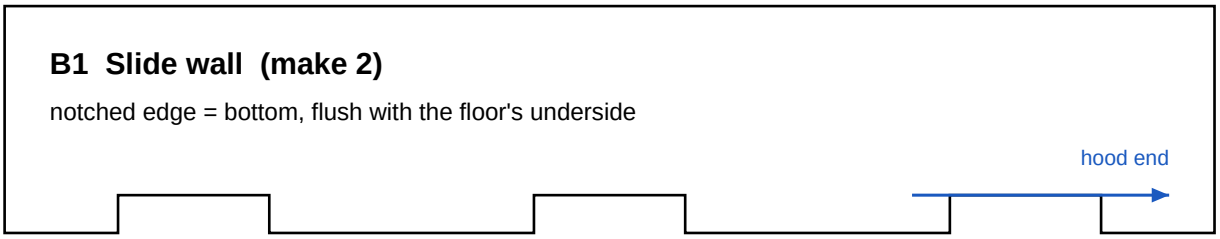


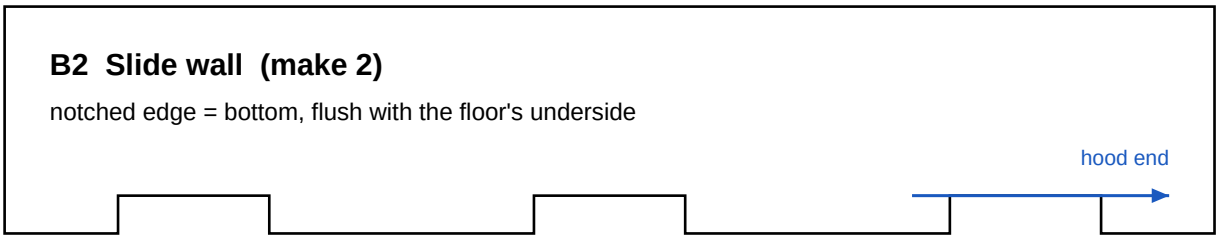

Card: 63 x 88 mm (sleeved ~66 x 91). Channel 70 mm. Cameras 130 mm from the window, centred where the card stops against the gate.

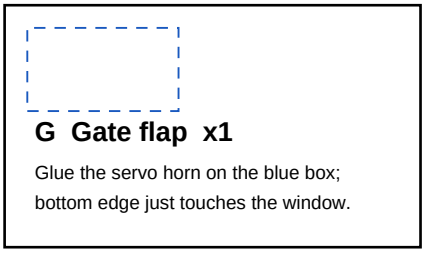
# SortBuddy scan station: slide, gate flap, joiners

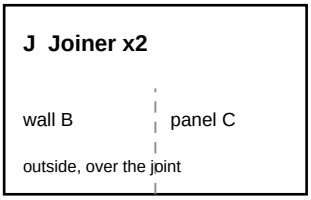


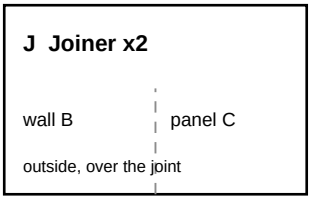


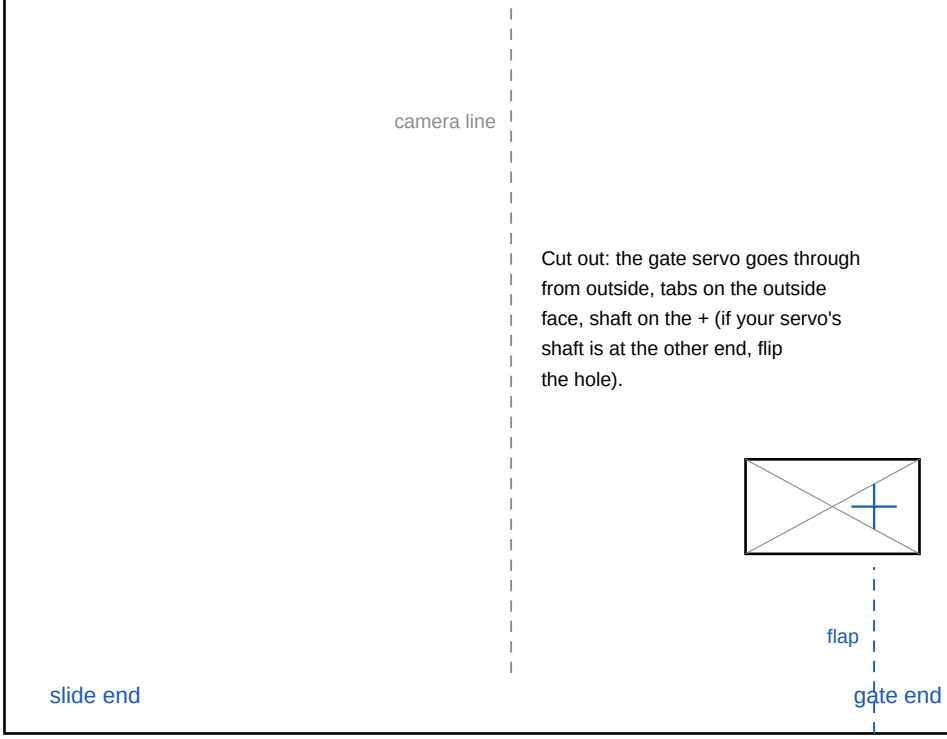



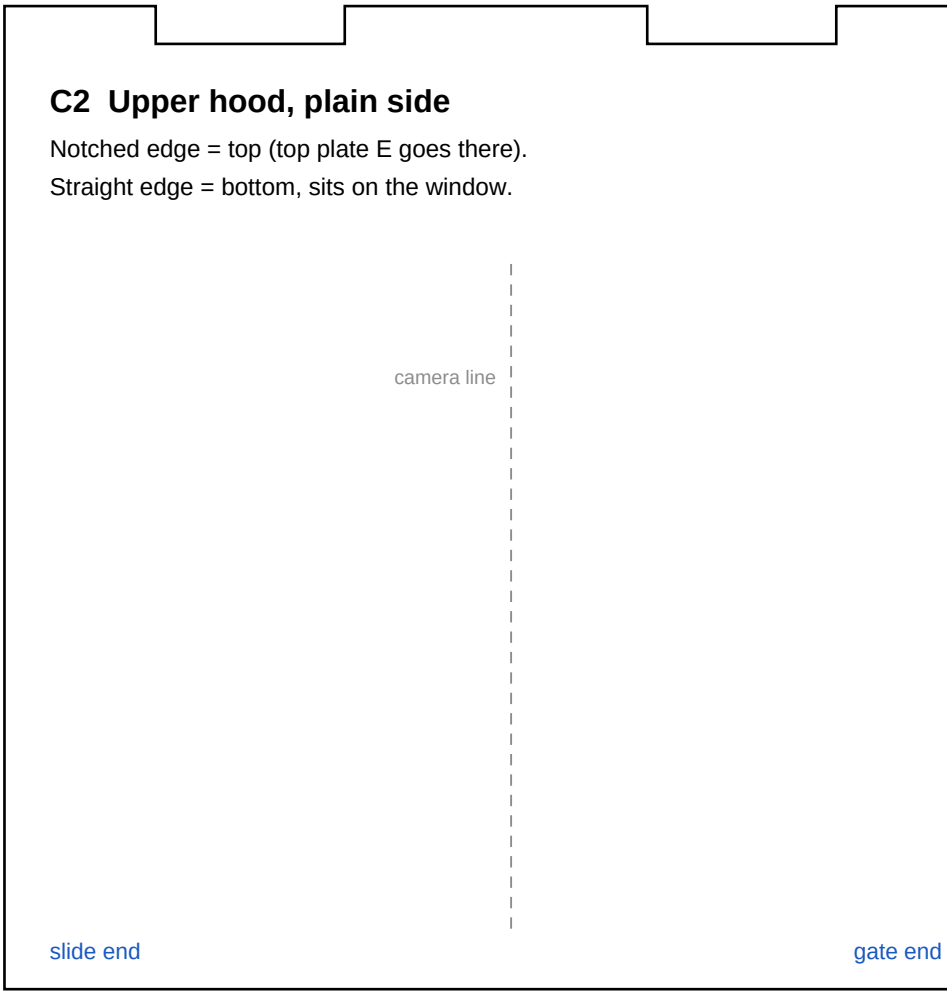
## C1 Upper hood, GATE side

Notched edge = top (top plate E goes there).
Straight edge = bottom, sits on the window.



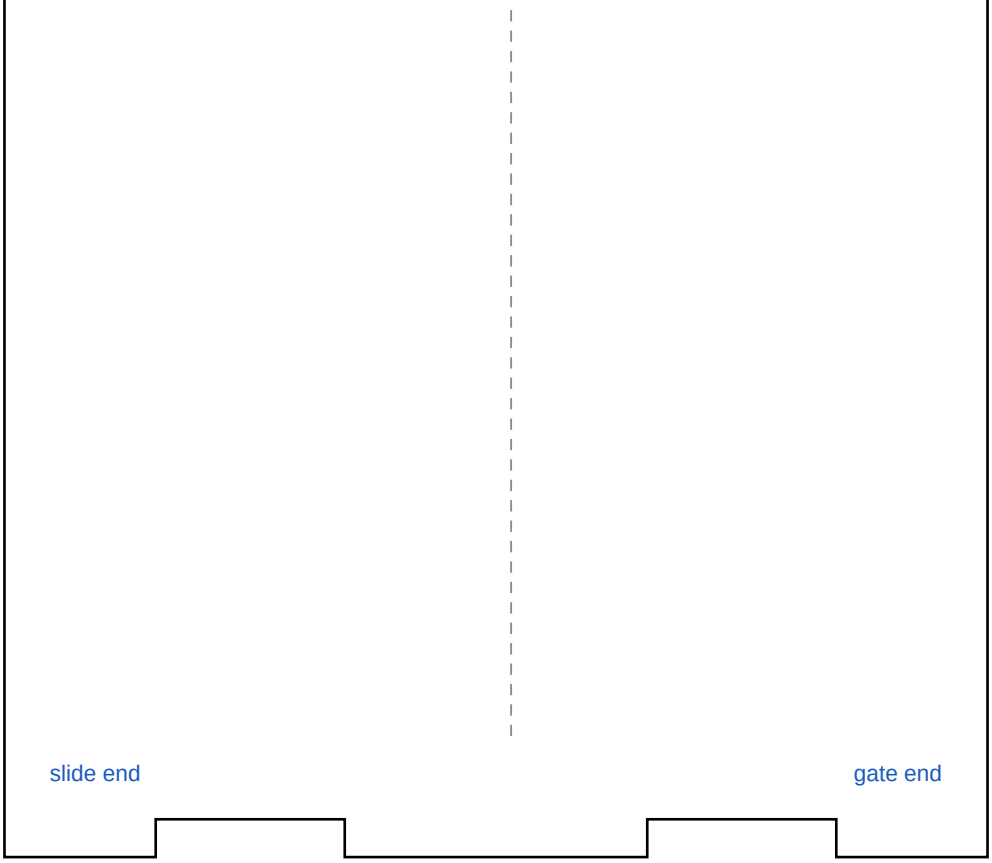

## C2 Upper hood, plain side

Notched edge = top (top plate E goes there).
Straight edge = bottom, sits on the window.

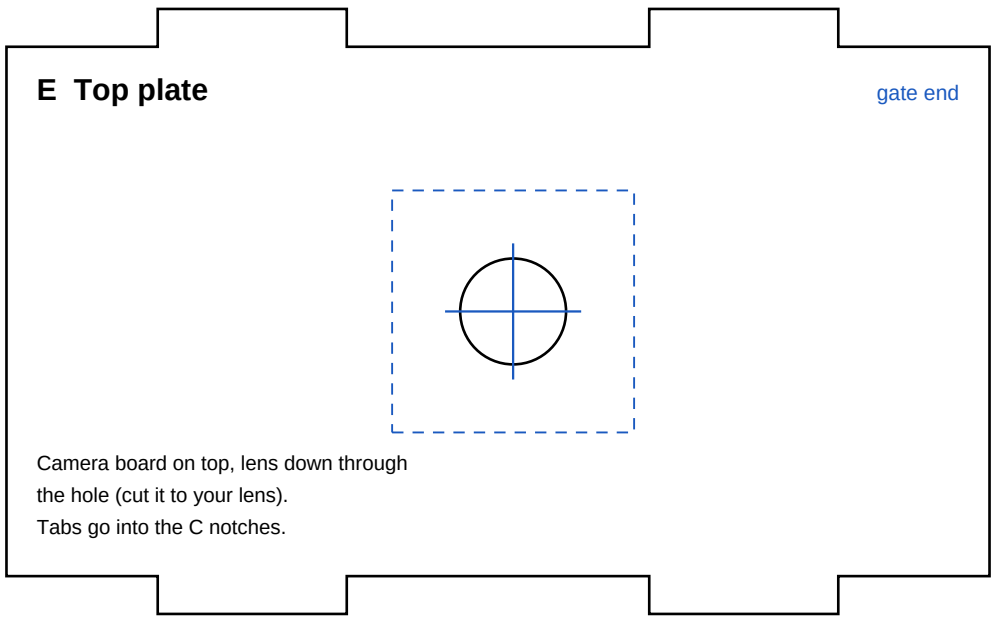


## D1 Lower hood panel (make 2)

Straight edge = top, the window rests on it.
Notched edge = bottom (bottom plate F goes there).




## E Top plate




Camera board on top, lens down through the hole (cut it to your lens).
Tabs go into the C notches.

## D2  Lower hood panel  (make 2)

Straight edge = top, the window rests on it.

Notched edge = bottom (bottom plate F goes there).

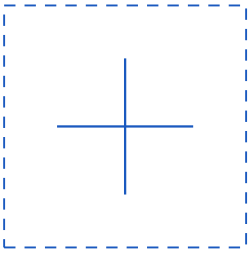


slide end

gate end

## F  Bottom plate

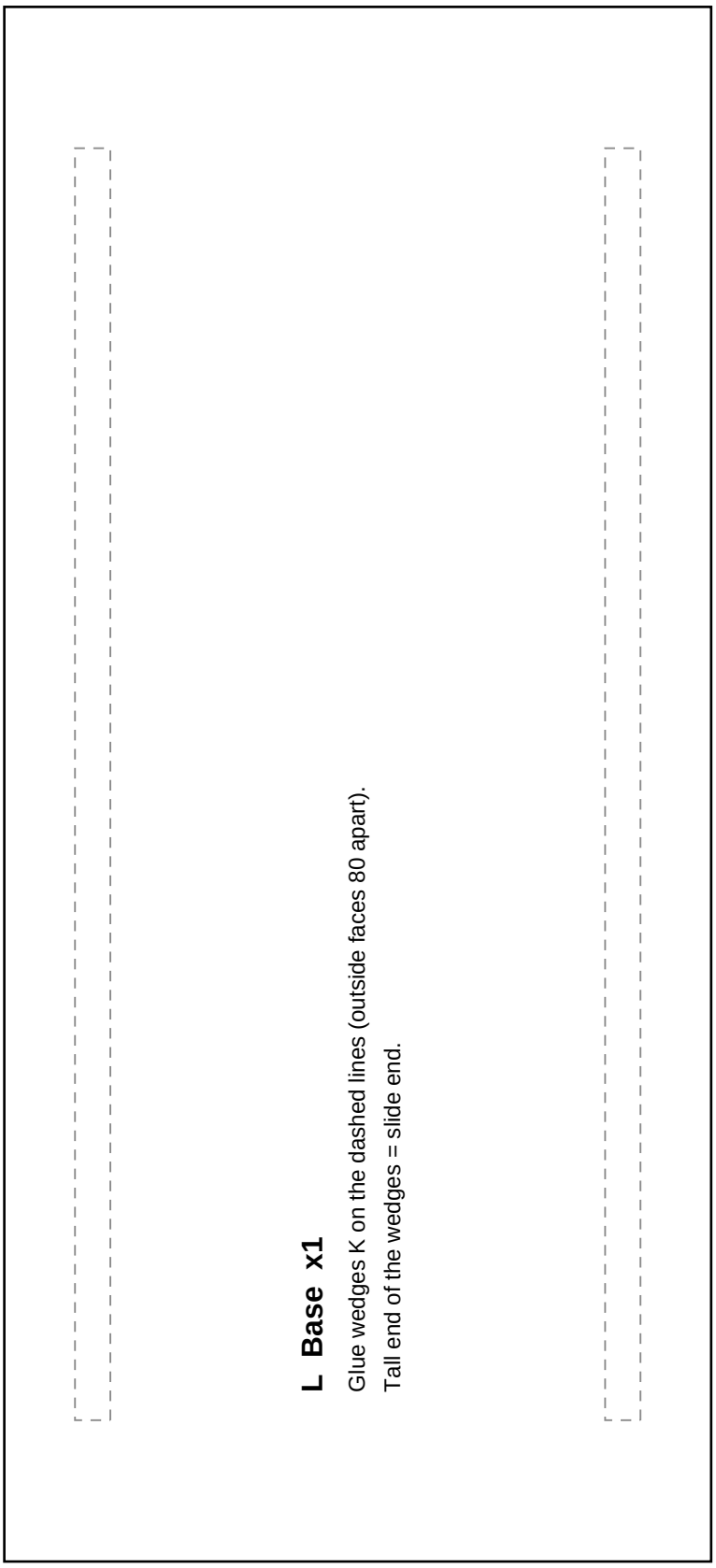
gate end



Camera board inside, on this square,
lens pointing up at the window.
Tabs go into the D notches.

# SortBuddy scan station: base and splices






**H Splice (make 4)**

upper panel C

window edge

lower panel D

glue on the OUTSIDE

**H Splice (make 4)**

upper panel C

window edge

lower panel D

glue on the OUTSIDE

**H Splice (make 4)**

upper panel C

window edge

lower panel D

glue on the OUTSIDE

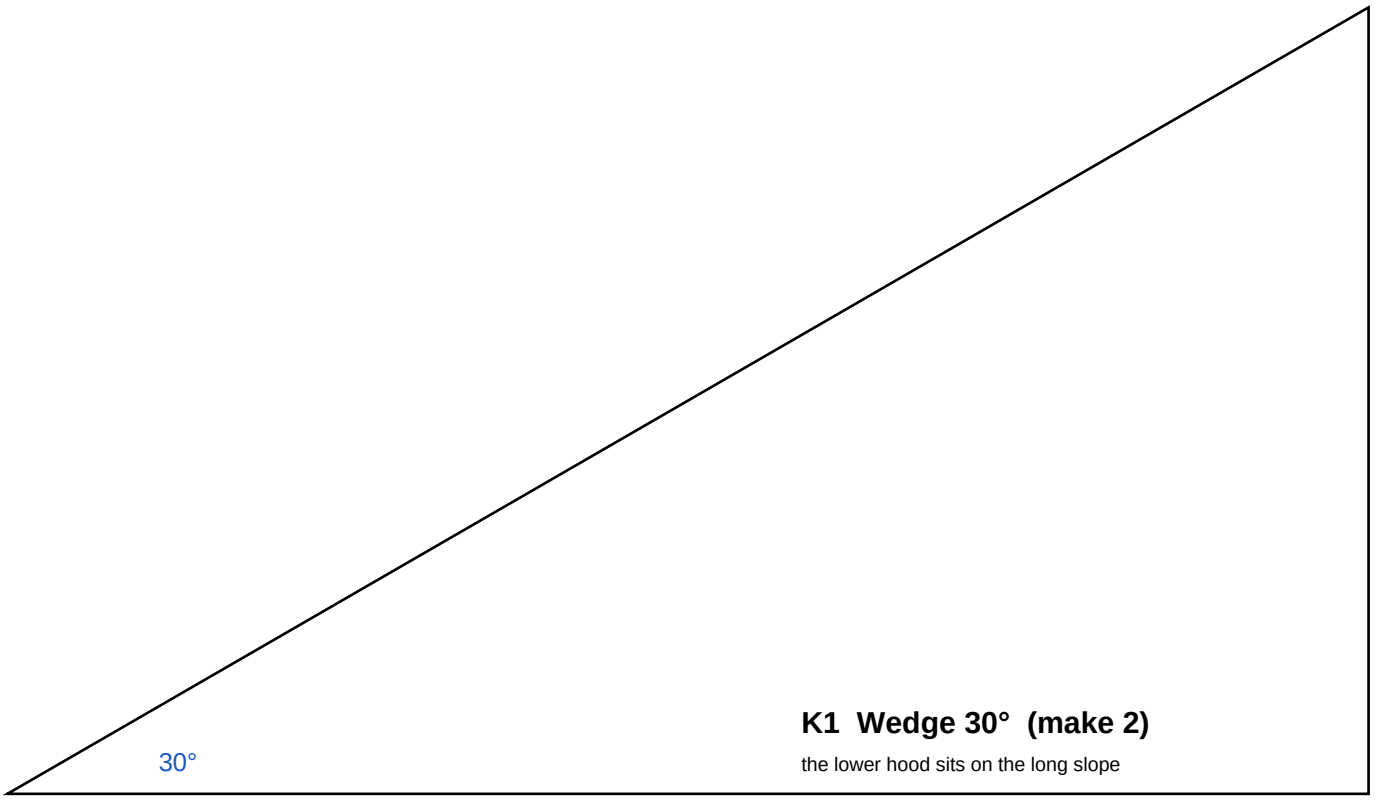
**H Splice (make 4)**

upper panel C

window edge

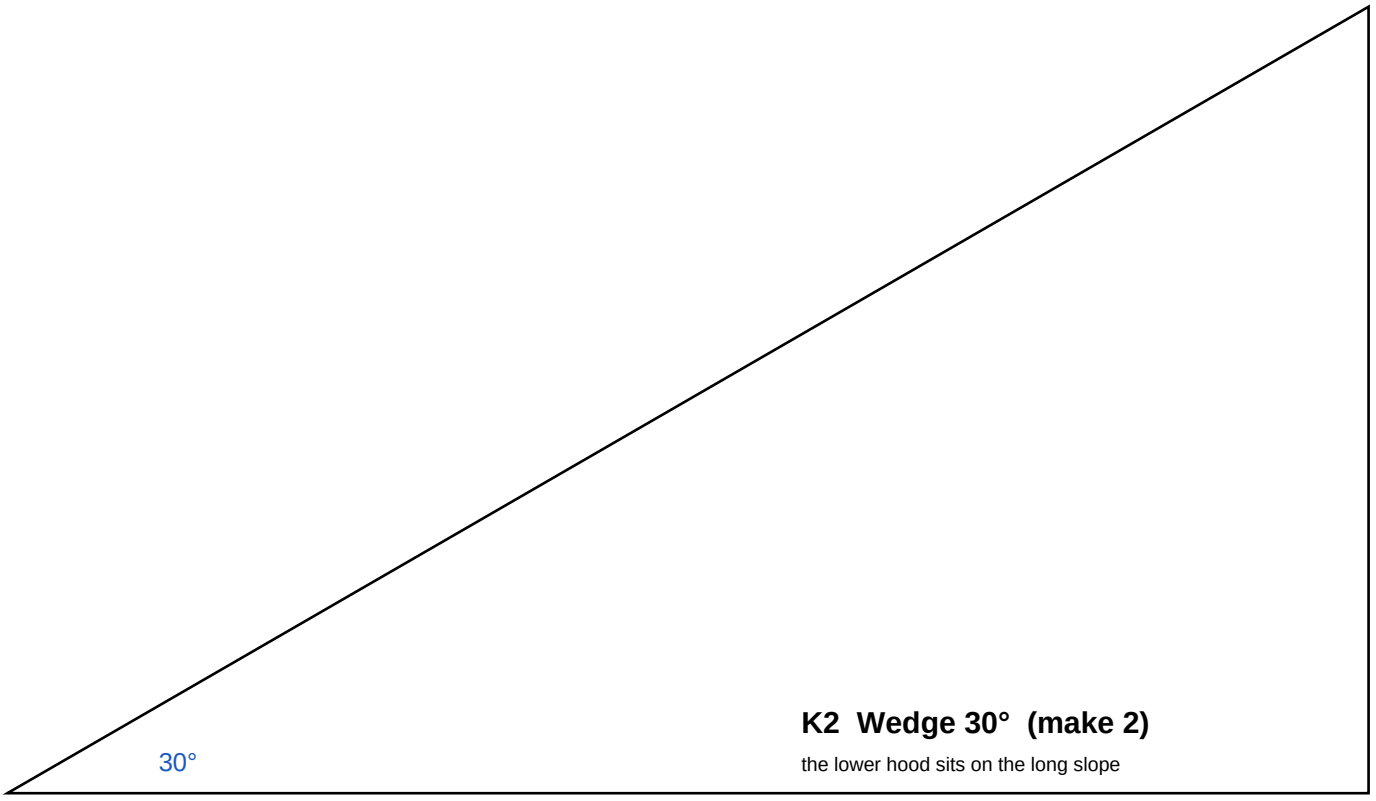
lower panel D

glue on the OUTSIDE




**K1 Wedge 30° (make 2)**

the lower hood sits on the long slope




**K2 Wedge 30° (make 2)**

the lower hood sits on the long slope

gate

## ACRYLIC window

80 x 130 from 2 mm acrylic

Lay the acrylic on this
and mark it. Score 10-15x
with a knife, snap over
the table edge.

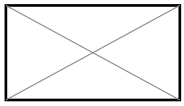
slide floor ends here

slide end

## M Sorter flap (later)

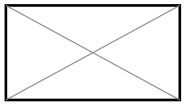
Cards land here after the gate.
Servo horn under the blue line:
tilt left = YES, right = NO,
level = straight on to ?

## N Servo holder (x2, later)



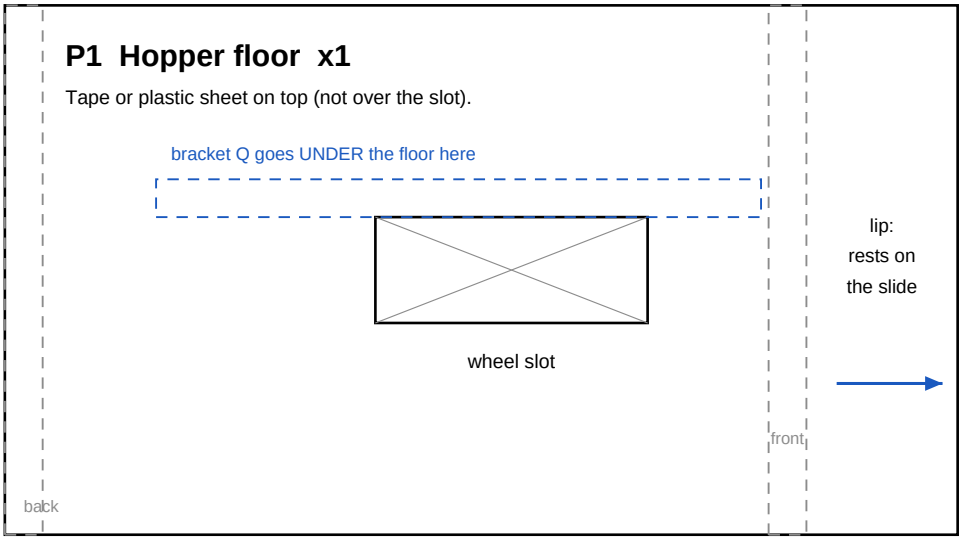
two layers glued together

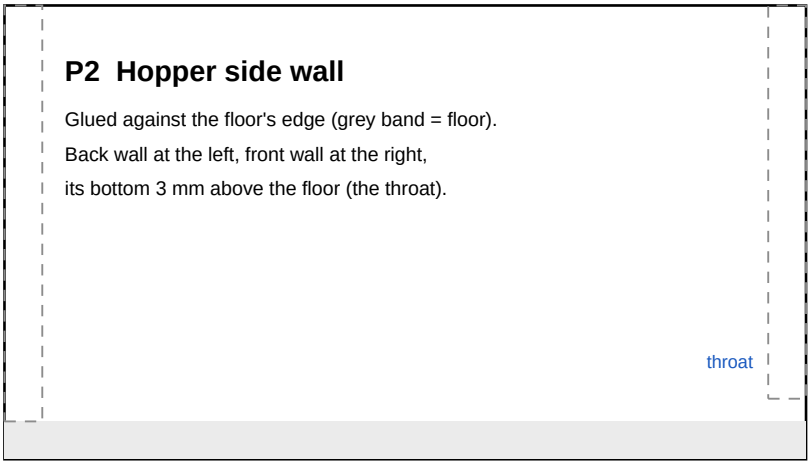
## N Servo holder (x2, later)

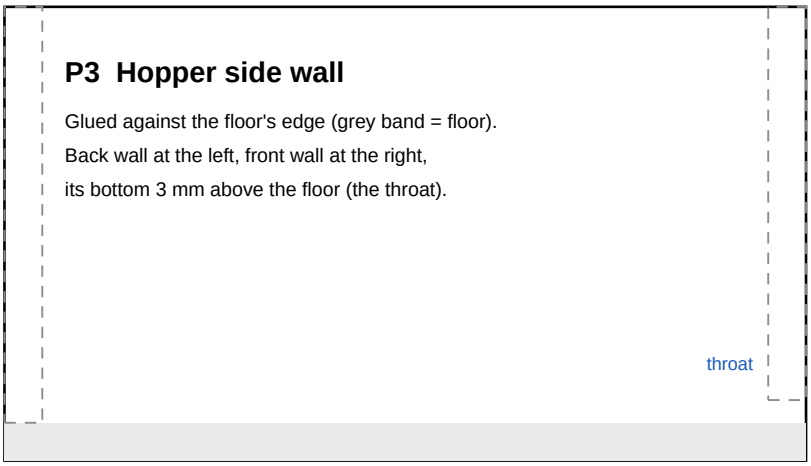


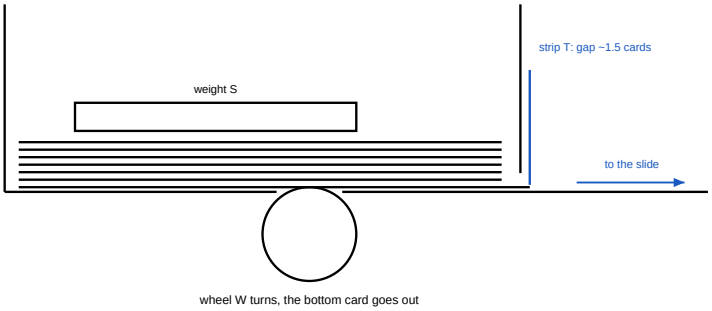
two layers glued together

# SortBuddy scan station: hopper floor and side walls




Side view, not to scale. The strip lets one card under and holds the next one back.

# SortBuddy scan station: hopper walls, wheel, motor bracket



## P4 Back wall

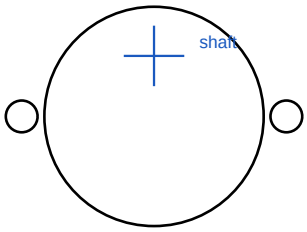
stands on the floor,
between the side walls

## P5 Front wall

bottom edge 3 mm above the floor

slots: M3 screws for strip T

## Q Motor bracket x1




Top edge under the floor (blue line on P1).
Motor's flat front on THIS side, body
through the big hole. M3 screws in the tabs.

## W Wheel

make 2,
glue together

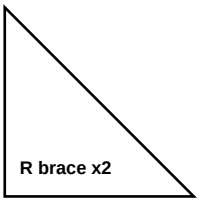
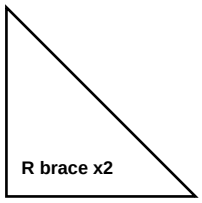
## W Wheel

make 2,
glue together

## S Stack weight

Sits on top of the cards
so the last few still
press on the wheel.
Tape coins on top
(about 30-50 g).

## T Throat strip (PLASTIC)

file-folder sheet or acrylic offcut;
screwed on the front wall's outside.

Bottom edge = the gap: ~1.5 cards.